SYDNEY GRAMMAR SCHOOL

# 2019
# TRIAL EXAMINATION

# BIOLOGY

## Form VI

**STRUCTURE OF PAPER**

**SECTION I - 20 marks** (pages 3-9)

- Attempt Questions 1- 20
- Allow about 35 minutes for this section

**SECTION II – 80 marks** (pages 11-34)

- Attempt Questions 21- 36
- Allow about 2 hours and 25 minutes for this section

**EXAMINATION**

| | |
|---|---|
| DATE: | **Tuesday 20th August 8.40 AM** |
| DURATION: | 3 hours + 5 minutes reading |
| MARKS: | 100 |

**CHECKLIST**

Each boy should have the following:

- 1 Examination Paper
- 1 Multiple-Choice Answer Sheet

**GENERAL EXAM INSTRUCTIONS**

- 5 minutes reading time and 3 hours working time
- Write using black pen and draw diagrams using pencil
- For questions in Section II, show all relevant working for questions involving calculations
- NESA approved calculators may be used
- Remove the centre staple and hand in all parts of the paper in a neat bundle.
- WRITE YOUR **CANDIDATE NUMBER** IN THE SPACE PROVIDED AT THE TOP OF EACH SEPARATE SECTION IN SECTION II.

| CLASS NUMBER | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| Class | 6BL201 | 6BL202 | 6BL203 | 6BL204 |
| Master Initials | DBD | HCKM | HCKM | GPW |

Authors: ACR, DBD, HCKM, GPW, Master in charge: HCKM

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## SECTION I : MULTIPLE CHOICE (20 marks)

Attempt ALL Questions
Use the Multiple-Choice Answer Sheet.

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**1** What name is given to a molecule that triggers an immune response?

(A) Toxin
(B) Antibody
(C) Antigen
(D) B-lymphocyte

**2** Where are phagocytes and lymphocytes produced?

(A) Bone marrow
(B) Lymph nodes
(C) Hypothalamus
(D) Thymus gland

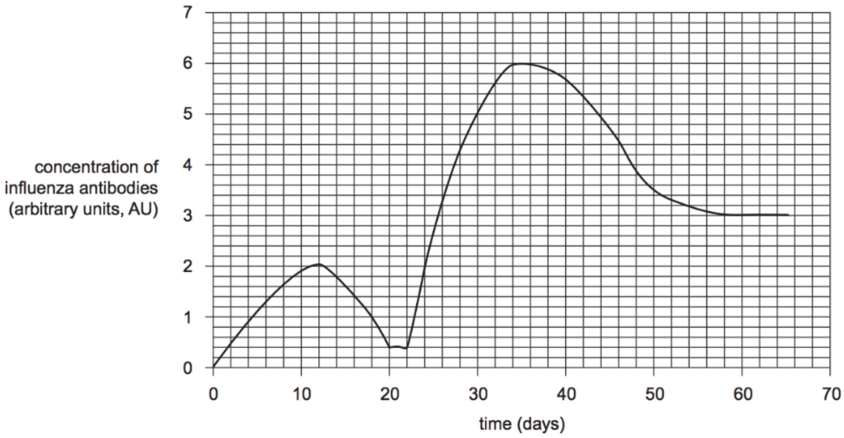
**3** What substance is secreted by T-helper cells to activate selected B-lymphocytes?

(A) Antibodies
(B) Cytokines
(C) Hormones
(D) Histamines

**4** Which of the following is the main barrier to entry of infectious agents into the human body?

(A) Cilia
(B) Skin
(C) Commensal bacteria
(D) Mucus membranes

**5** A daily blood sample was obtained from an individual who received a single vaccination against a particular strain of the influenza virus. The individual had no prior exposure to this strain of influenza. The graph below shows the concentration of antibodies present in the individual's blood for this strain of influenza over a period of 65 days.




Which one of the following conclusions can be made using this data?

(A) Memory B cells were activated by exposure to the same strain of the influenza virus on day 22.

(B) B plasma cells specific to this strain of influenza were most numerous on day 12.

(C) Herd immunity to this particular strain of influenza was achieved by day 55.

(D) The vaccination containing weakened influenza antigens occurred on day 10.

**6** During meiosis, recombinant chromatids arise from:

(A) the alignment of chromosomes at metaphase II.

(B) the reciprocal exchange of DNA between homologous chromosomes during prophase I.

(C) the exchange of DNA between non-homologous chromosomes during prophase I.

(D) the reciprocal exchange of DNA between sister chromatids during prophase.

**7** If a segment of mRNA molecule has a sequence 5' GUACCGAUCG 3', which of the following could have been the template DNA molecule?

(A) 5' GCUAGCCAUG 3'

(B) 5' CGATCGGTAC 3'

(C) 5' GUACCGAUCG 3'

(D) 5' CATGGCTAGC 3'

**8** Which term is used to describe the cell formed after fertilisation?

(A) Embryo

(B) Foetus

(C) Zygote

(D) Blastocyst

**9** What is the role of oxytocin in the female reproductive cycle?

(A) Initiates menstruation.

(B) Thickens the uterine lining.

(C) Stimulates the release of an egg.

(D) Stimulates contractions in the uterus.

**10** Which of the following statements about phenotype expression is the most correct?

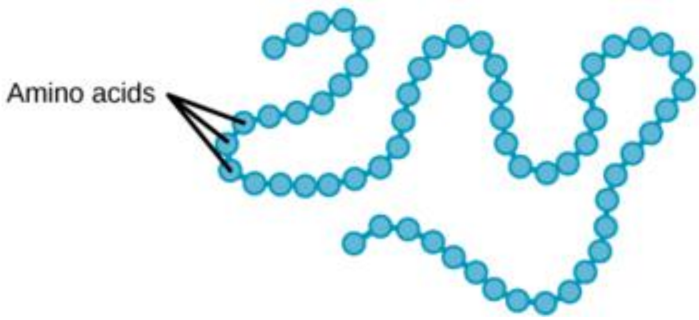
(A) All genes are responsible for the phenotype.

(B) Both genotype and environment affect the phenotype.

(C) Only the dominant genes affect the phenotype.

(D) Genotype indirectly affects phenotype by triggering other systems.

**11** The basic structure of which molecule is represented by the diagram below?




(A) DNA

(B) Protein

(C) Glycogen

(D) Nucleotide

**12** Analyse the Punnett Square below.

| | b | b |
|---|---|---|
| B | Bb | Bb |
| b | bb | bb |

Which statement is correct about the phenotype of the individuals in this cross?

(A) The offspring and the parents all have different phenotypes.

(B) All offspring will have the same phenotype.

(C) One of the parents has the recessive phenotype.

(D) Both parents have the recessive phenotype.

**13** A single nucleotide DNA change from an A to a T occurs and is copied during replication; is this change in DNA sequence necessarily a mutation?

(A) Yes, it is a change in the DNA sequence.

(B) Yes, but only if the nucleotide change occurs in the sex cells.

(C) Yes, but only if the nucleotide change occurs in the coding part of a gene.

(D) No, because A and T might be at the 3rd position of a codon and this might not change the amino acid.

**14** Cytotoxic T cells destroy infected cells by which of the following methods?

(A) Phagocytosis

(B) Secretion of antibodies.

(C) Injection of oxidising agents into the infected cell's cytoplasm.

(D) Release of chemicals that put holes in the target cell's membrane.

**15** DNA differs between prokaryotes and eukaryotes for which of these reasons?

(A) Prokaryotes have telomeres, and eukaryotes do not.

(B) Prokaryotic DNA is single stranded, whereas eukaryotic DNA is double stranded.

(C) Prokaryotic DNA is circular, whereas eukaryotic DNA is only linear.

(D) Prokaryote DNA is not associated with histones, whereas eukaryotic DNA is wrapped around histones.

**16** The trait for "male-pattern baldness" is a recessive trait and the allele for this trait is encoded by the letter "b". Non-balding is encoded for by a dominant allele encoded by the letter "B". A survey of men over the age of 30 years old revealed that out of 500 men, 150 had male pattern baldness. Of the 350 men who had a full a full head of hair, 100 were homozygous. What is the frequency of the b allele in this population?

(A) 0.15

(B) 0.40

(C) 0.45

(D) 0.55

**17** How is the nervous impulse transferred across the synaptic gap between adjacent neurones?

(A) The active transport of ATP molecules.

(B) The release of hormones from the axon terminal.

(C) The diffusion of sodium ions across the synapse.

(D) The release of neurotransmitters from synaptic vesicles.

**18** In 1954, copper waste in the Finniss River killed numerous fish. This caused various species in the area to die out. However, one species, the black-banded rainbow fish, increased in numbers. The black-banded rainbow fish have modified gills that enable the fish to filter and remove the copper before it enters their body.

With respect to the black-banded rainbow fish, it is reasonable to conclude that

(A) a mutation occurred in their population in 1954.

(B) the ability of their gills to remove copper already existed in 1954.

(C) the high levels of copper in the water changed the structure of their modified gills.

(D) their genomes are identical with those of the other species of fish that existed in 1954.

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**19** The follwing table outlines two mutations:

| | DNA sequence |
|---|---|
| **Original DNA** | **AAT TTT GCG GTC** |
| Mutation 1 | AAA TTT GCG GTC |
| Mutation 2 | AAT ATT TGC GGT |

Which row below correctly identifies the type of mutation that is mutation 1 and mutation 2, respectively?

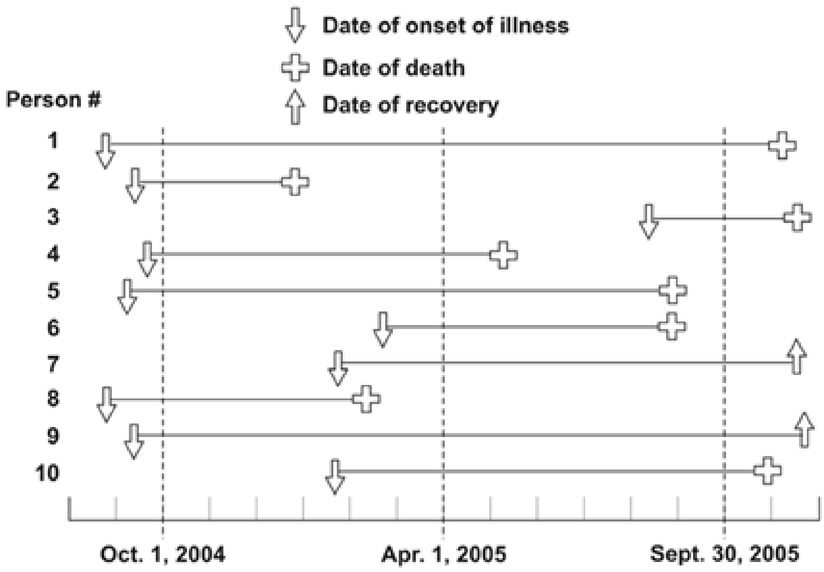
(A) Substitution Deletion

(B) Addition Deletion

(C) Substitution Insertion

(D) Addition Insertion

**20** The following diagram shows the history of a particular disease on a navy ship with a crew of 100 people. Only the data for individuals who caught the disease is shown.




From the information provided above, what is the incidence and prevalence of this disease on the ship for the time period 1st April to 30th September 2005?

| | Incidence (%) | Prevalence (%) |
|---|---|---|
| (A) | 0.2 | 2.0 |
| (B) | 1.1 | 8.2 |
| (C) | 8.0 | 1.2 |
| (D) | 12.5 | 80.0 |

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## SECTION IIA : SHORT AND LONG ANSWER (28 Marks)

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CANDIDATE NUMBER

Answer the questions in the spaces provided.
Show all relevant working in questions involving calculations.

**Question 21** (2 marks) **Marks**

Plants are a rich source of nutrients for many organisms, including bacteria, fungi and viruses. Although plants lack an immune system that is comparable to animals, they have evolved a number of passive and active defences to stop invading pathogens from causing significant damage.

Outline two responses of plants that protect them from an invading pathogen. Include at least one active defence response in your answer.

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**Question 22** (2 marks)

Immunity can be defined as the ability to resist infection.
A person can acquire immunity actively or passively. Complete the table to show some of the features of passive and active immunity. Place a tick in the box if the feature applies and a cross if it does not apply.

| | The person produces an immune response | The person produces memory cells | The immunity can be acquired naturally and artificially |
|---|---|---|---|
| Passive immunity | | | |
| Active immunity | | | |

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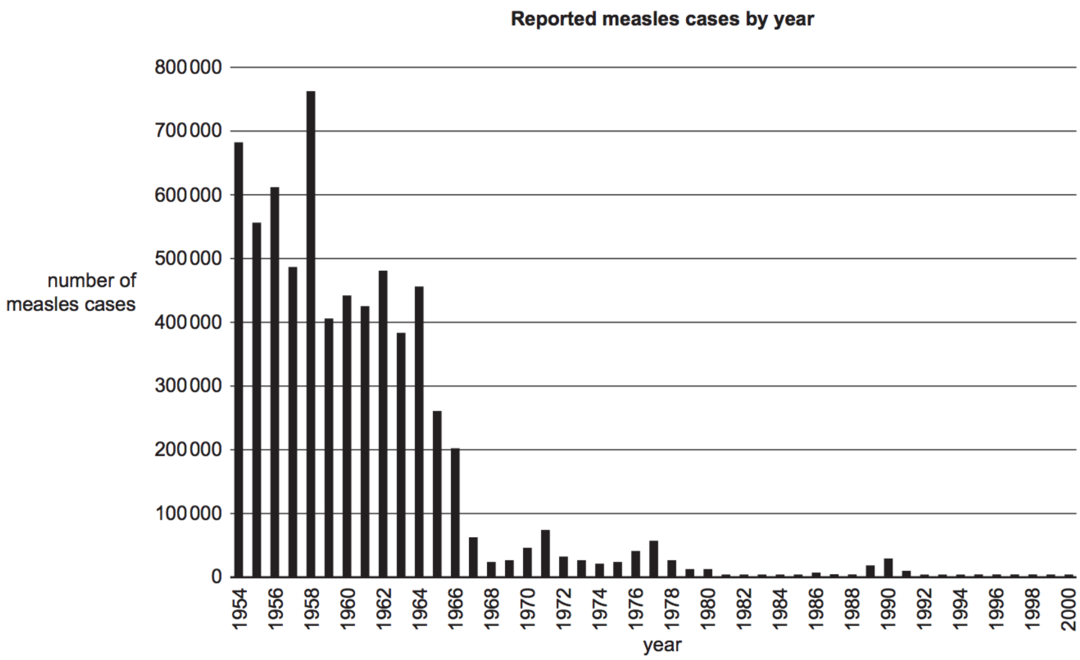
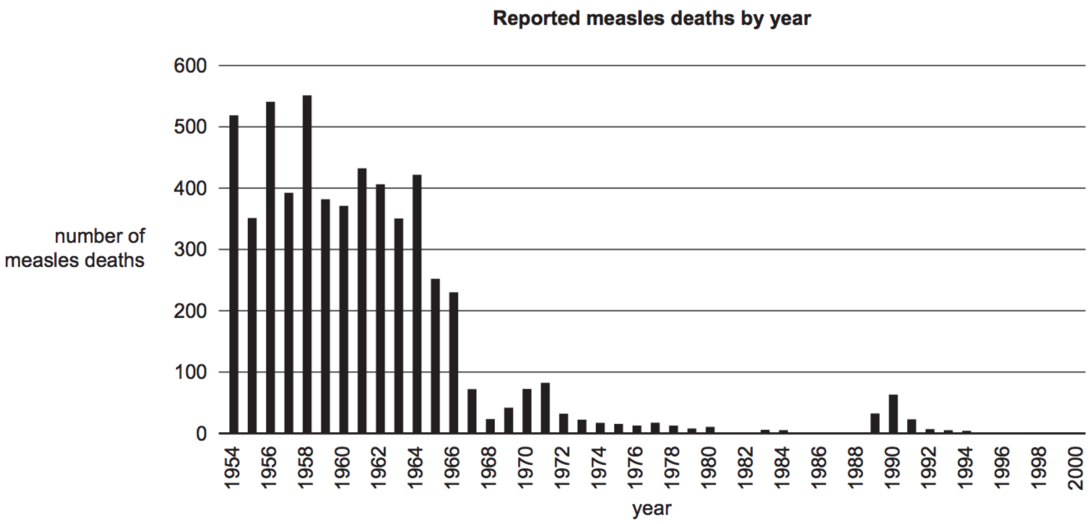
**Question 23** (4 marks) **Marks**

Measles is a highly infectious and dangerous disease. Young children and individuals with impaired immunity are especially susceptible to measles. Analyse the following graphs that show the number of people in the United States of America who were infected with measles during the period 1954–2000 and the number of people who died as a result of having measles during the same period.




**Question 23 continued on next page.**

**Question 23 continued.** **Marks**

(a) What trends can be observed when the two graphs are compared?

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(b) Controlling the number of measles' cases in a population relies on herd immunity. What is herd immunity and explain how it helps control the number of cases of this disease?

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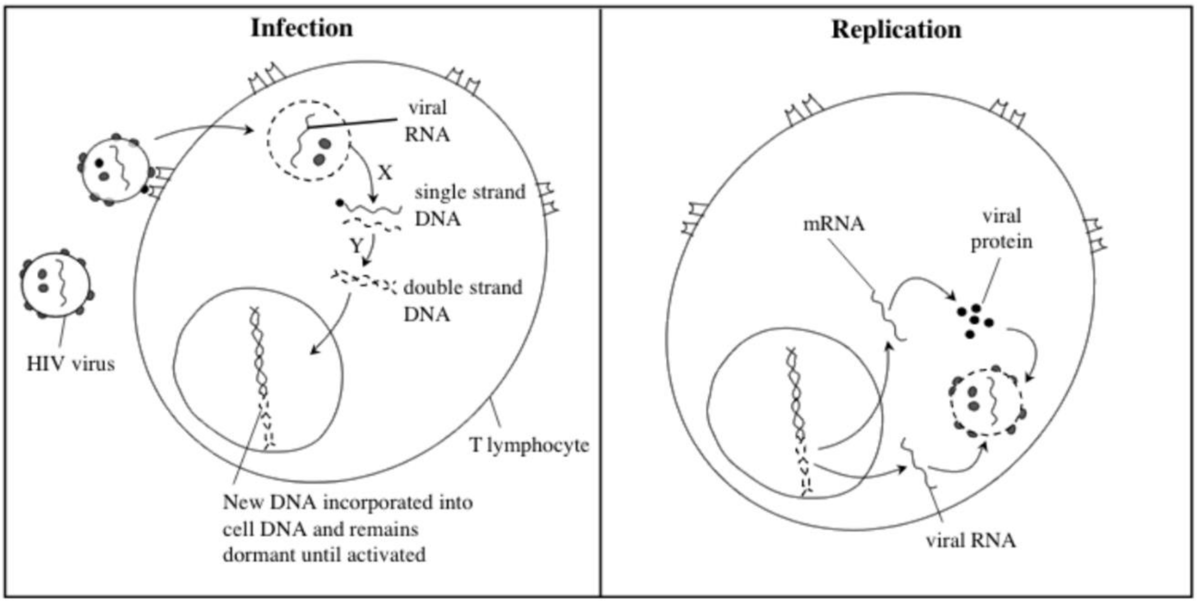
**Question 24** (11 marks) **Marks**

Infection with the human immuno-deficiency virus (HIV) can cause AIDS. The virus infects some types of T lymphocytes and replicates within them. Eventually, infection with HIV can lead to AIDS, in which the person may develop tumours and a number of infectious diseases.

(a) The mechanism of infection and replication is shown in the diagram below:




i. Suggest how the virus "recognises" the T lymphocytes.

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ii. What type of viral enzyme controls the processes labelled Y? Give evidence from the diagram to support your answer.

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**Question 24 continued.** **Marks**

iii. Describe TWO structural characteristics that could be used to distinguish between HIV and an infective prion.

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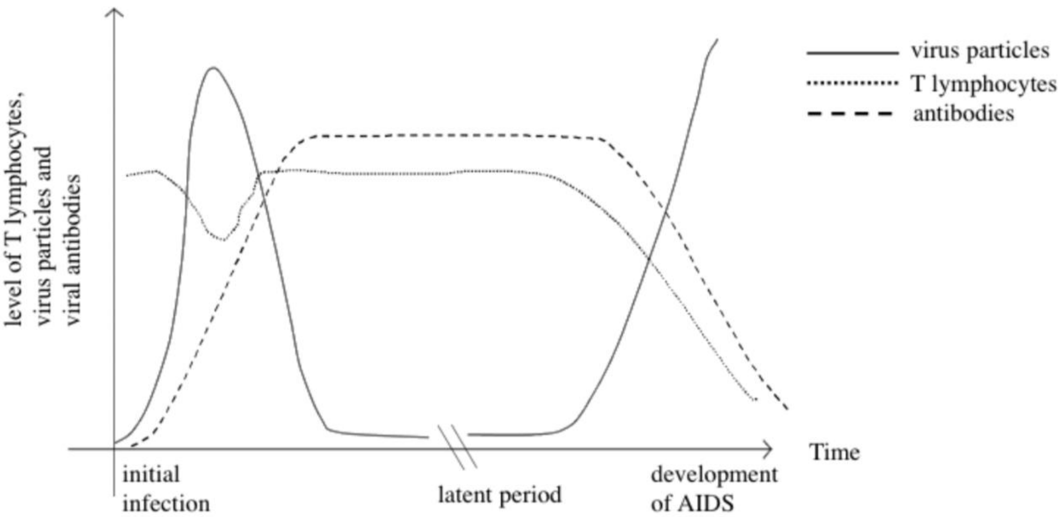
(b) The graph shows the changes in the levels of HIV particles, HIV antibodies and T lymphocytes in the blood during the different stages of HIV infection.




i. Describe the production and function of antibodies in the initial HIV infection.

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**Question 24 continued.** **Marks**

ii. With reference to the graph on the previous page, explain why, in the final stage of HIV infection, a person may develop tumours and a range of other diseases.

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**Question 25** (9 marks) **Marks**

The World Health Organisation (WHO) collects data on malaria, which is a disease caused by a Plasmodium parasite. For 2017, the WHO used the data shown in the table below to estimate:

- the total number of people with the disease in each country
- the total number of deaths from malaria in each country

| Country | Number of people with malaria per 100,000 | Number of deaths from malaria per 100,000 |
|---|---|---|
| Niger | 21,360 | 135 |
| Mali | 16,490 | 96 |
| Uganda | 19,230 | 55 |
| Cambodia | 470 | 8 |
| Indonesia | 330 | 3 |

(a) Give one reason why it is necessary to express the number of people as 'per 100,000'

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(b) Calculate the percentage of those with malaria in Cambodia who survived the disease. Show all working.

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**Question 25 continued on next page.**

**Question 25 continued.**

(c) For malaria or another named infectious disease, evaluate strategies that can be employed to prevent or limit the spread of the disease locally, regionally and globally. **6**

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**Question 25 continued.**

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## SECTION IIB : SHORT AND LONG ANSWER (27 Marks)

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CANDIDATE NUMBER

Answer the questions in the spaces provided.
Show all relevant working in questions involving calculations.

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**Question 26** (4 marks) **Marks**

Look at the diagram below showing the offspring of two pure breeding camellias.


(a) What type of inheritance pattern do these results indicate?

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(b) Use a punnett square to determine the genotype and phenotype ratios of the offspring from a cross between two of the **red and white stripped** flowers.

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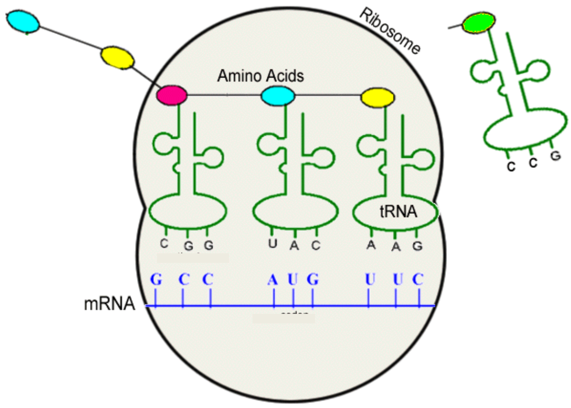
**Question 27** (4 marks) **Marks**

The diagram below shows a model of the translation process during polypeptide synthesis.




With reference to the diagram, assess the importance of mRNA and tRNA in this process. **4**

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**Question 28** (3 marks) **Marks**

In sexual reproduction, fertilisation can be either external or internal. Explain how the relationship between the number of gametes produced and the type of fertilisation ensures the continuity of the species.

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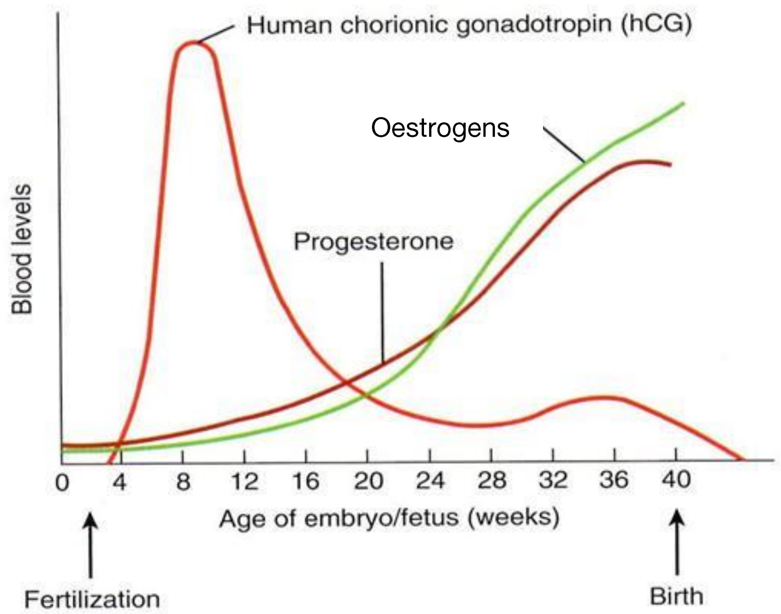
**Question 29** (4 marks) **Marks**

Study the graph below which shows the relative changes in some hormones in the blood during pregnancy.




(a) From this graph, identify the hormone, whose presence in the blood confirms that the woman is pregnant.

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(b) Explain the role that changing levels of hCG and progesterone play during pregnancy.

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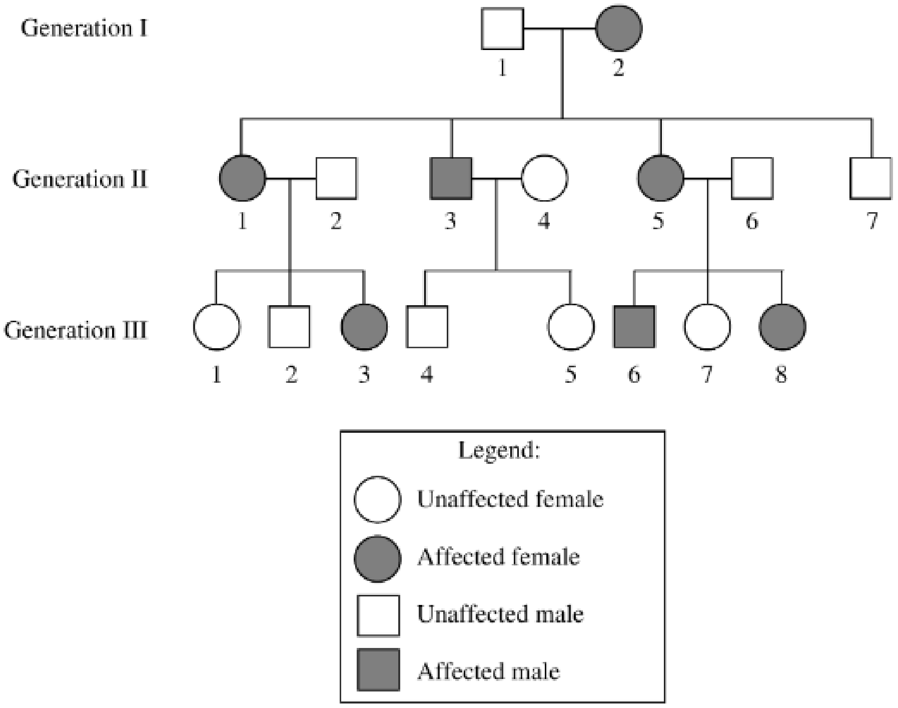
**Question 30** (12 marks) **Marks**

The following pedigree chart shows the inheritance pattern for a rare genetic disease.




(a) Analyse the pedigree to determine the possible mode(s) of inheritance for this disease. Justify your decision with evidence from the pedigree.

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(b) What is the probability that the next child of individuals II 3 and II 4 will have the disease?

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**Question 30 continued on next page.**

**Question 30 continued.** **Marks**

(c) Critically evaluate the use of pedigrees and genetic technologies for accurately determining and predicting inheritance patterns, and genetic relationships in populations. **8**

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**Question 30 continued.** **Marks**

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## SECTION IIC : SHORT AND LONG ANSWER (25 Marks)

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Answer the questions in the spaces provided.
Show all relevant working in questions involving calculations.

**Question 31** (3 marks)

For a named transgenic organism, outline the techniques used to:

- remove the desired gene from species 1
- make copies of the desired gene
- insert the desired gene into the genome of species 2

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**Question 32** (2 marks)

A pea plant is homozygous recessive for gene A and heterozygous for gene B. These two genes are located on the same chromosome. It is also homozygous dominant for gene E, which is located on a separate chromosome. Draw a model to show the alternative ways this plant's chromosomes could be arranged during metaphase of meiosis I. Assume crossing over does not occur. **Show all the genes and alleles.**

**Question 33** (6 marks) **Marks**

The southern brush-tailed rock wallaby (*Petrogale penicillata*) is threatened with extinction in Victoria.

Since European settlement, it has suffered from hunting for its fur, clearing of habitat and predation by foxes. Early in 2012, the population of brush-tailed rock wallabies in the Grampians National Park numbered only four individuals. Eighteen brush-tailed rock wallabies were released into the Grampians in late 2012. The wallabies had been bred in captivity in zoos and nature reserves.

(a) Define what is meant by a gene pool?

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(b) Explain why it is important to analyse population genetics data for the conservation management of the brush-tailed wallabies.

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**Question 33 continued on next page.**

**Question 33 continued.** **Marks**

(c) The gene pool of a small population of wallabies could be affected by natural mechanisms. Explain how genetic drift and gene flow could possibly result in a change in the frequency of alleles in the population.

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**Question 34** (4 marks)

Evaluate the effect on biodiversity of using transgenic organisms and whole animal cloning in agriculture.

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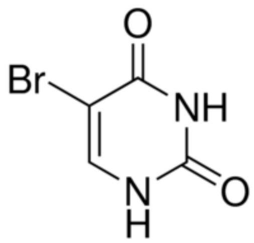
**Question 35** (5 marks) **Marks**

Complete the table below to show how different types of mutagens can affect genetic material in cells.

| **Type of mutagen** | **Example** | **Explain how it affects the genetic material** |
|---|---|---|
| Electromagnetic radiation | | Cross-links thymine bases, distorting the DNA, so these two bases no longer pair with their adenine partners. |
| | 5-bromouracil  | |
| Naturally occurring | | |

**5**

**Question 36** (5 marks) **Marks**

Construct a negative feedback loop to show the specific mechanisms that occur to bring core body temperature back to its set point when it is too high or too low in an endotherm.

In your diagram label the:

- stimuli
- receptor
- control centre
- effectors
- responses
- negative feedback

**5**

END OF EXAMINATION